

EarthBridge: A Solution for 4th Multi-modal Aerial View Image Challenge Translation Track

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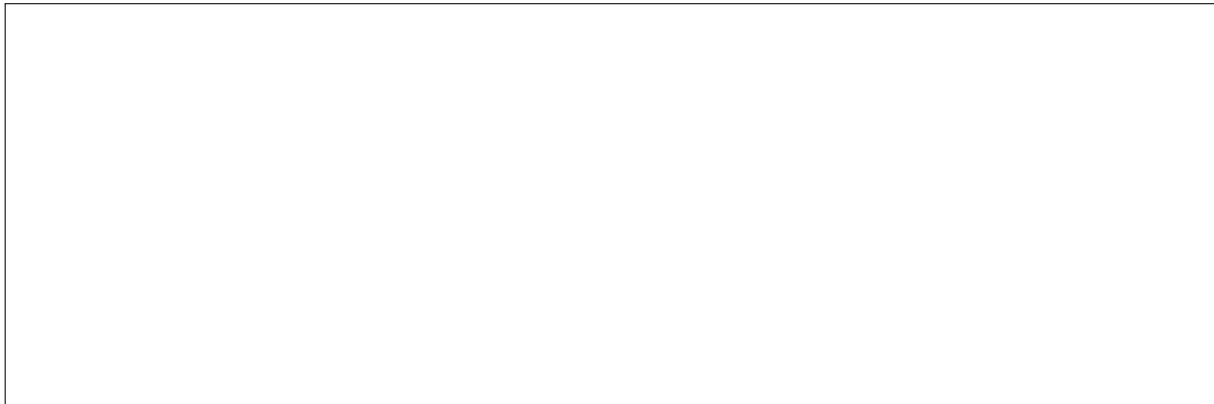


Figure 1. Teaser: The proposed EarthBridge framework for multi-modal aerial-view image translation across Electro-Optical (EO), Infrared (IR), and Synthetic Aperture Radar (SAR) modalities. Our method accurately bridges the domain gap, preserving structural integrity while synthesizing high-fidelity target textures.

Abstract

001 Cross-modal image-to-image translation among Electro-
002 Optical (EO), Infrared (IR), and Synthetic Aperture Radar
003 (SAR) sensors is essential for comprehensive multi-modal
004 aerial-view analysis. However, translating between these
005 modalities is notoriously difficult due to their distinct elec-
006 tromagnetic signatures and geometric characteristics. This
007 paper presents **EarthBridge**, a high-fidelity translation
008 framework developed for the 4th Multi-modal Aerial View
009 Image Challenge – Translation (MAVIC-T). **EarthBridge**
010 is founded on **Diffusion Bridge Implicit Models (DBIM)**,
011 which we generalize using non-Markovian bridge pro-
012 cesses to enable efficient, high-quality deterministic sam-
013 pling in as few as **[TODO: 40]** steps. Our method em-
014 ploys a channel-concatenated UNet denoiser trained with
015 Karras-weighted bridge scalings and a specialized “boot-
016 ing noise” initialization to handle the inherent ambiguity
017 in cross-modal mappings. To accommodate high-resolution
018 1024×1024 imagery, we implement a multi-stage training
019 protocol involving random-crop pre-training followed by

full-resolution fine-tuning. We evaluate **EarthBridge** across
all four challenge tasks (SAR→EO, SAR→RGB, SAR→IR,
RGB→IR), achieving superior spatial detail and spectral
accuracy. Our solution achieved a composite score of
[TODO: 0.XXX], securing a **[TODO: Rank]** position on the
MAVIC-T leaderboard.

1. Introduction

The Multi-modal Aerial View Image Challenge for cross-
domain image translation [1, 19, 20] is a leading chal-
lenge held in conjunction with the Perception Beyond the
Visible Spectrum (PBVS) workshop from 2023 to 2025.
Data collected across multiple sensing modalities—such as
Electro-Optical (EO), Infrared (IR), and Synthetic Aper-
ture Radar (SAR)—provide complementary perspectives on
the same scene. However, data coverage limitations and
modality-specific constraints often hinder the full utiliza-
tion of these diverse sources. For instance, while SAR
sensors offer all-weather capabilities and can penetrate at-
mospheric obstructions, their imagery is complex to inter-

pret, and publicly available datasets are scarce. Conversely, widely available EO imagery is limited by lighting conditions and atmospheric visibility. Domain translation seeks to bridge this gap by converting one modality into another, enabling cross-modal data synthesis and enhancing sensor interoperability. By focusing on translation tasks such as RGB→IR, SAR→EO, SAR→IR, and SAR→RGB, the MAVIC-T challenge aims to bolster existing data volume, increase data diversity across regions, and serve as a platform for advancing conditioned image synthesis techniques, ultimately fostering the development of more adaptable and robust models for multi-modal image analysis.

SAR images possess unique attributes that present challenges for both human observers and vision artificial intelligence (AI) models to interpret, owing to their electromagnetic (EM) characteristics [10–13, 21].

In this paper, we propose **EarthBridge**, a diffusion-based framework for cross-modal aerial image translation built on Diffusion Bridge Implicit Models (DBIM) [43]. Our approach generalizes the diffusion bridge framework with non-Markovian bridge processes, enabling efficient deterministic sampling. By conditioning the bridge process on the source image via channel concatenation and training a UNet denoiser under Karras-weighted bridge scalings [15], EarthBridge achieves high-fidelity translation across disparate modalities. We utilize pixel-level modeling throughout to preserve the full spectral and spatial resolution of the remote sensing data.

Our findings indicate that the deterministic DBIM sampling significantly improves inference speed, achieving high-quality results in as few as [TODO: 40] sampling steps. Qualitatively, EarthBridge demonstrates a superior ability to reconstruct complex urban layouts and fine-grained textures in challenging tasks such as SAR-to-RGB, effectively bridging the domain gap. Quantitatively, our solution achieves a composite task score of [TODO: 0.XXX] for the SAR→EO task and [TODO: 0.XXX] overall, ultimately placing [TODO: Rank] in the MAVIC-T challenge leaderboard.

Our main contributions are summarized as follows:

- We propose **EarthBridge**, a novel framework for paired cross-modal aerial image translation based on Diffusion Bridge Implicit Models (DBIM), which generalizes the bridge process with non-Markovian dynamics and enables efficient deterministic sampling.
- We develop a specialized training protocol using Karras-weighted bridge scalings and pixel-level modeling, optimized for the high dynamic range and resolution requirements of multimodal remote sensing data.
- We introduce a multi-stage training strategy involving random crop pre-training and full-resolution fine-tuning to effectively handle 1024×1024 translation tasks within practical memory constraints.

- We conduct comprehensive experiments on all four MAVIC-T tasks (SAR→EO, SAR→RGB, SAR→IR, RGB→IR), reporting state-of-the-art results and providing detailed quantitative and qualitative analysis.

2. Related Work

Translating across remote sensing (RS) modalities, such as SAR to optical (S2O), is critical for multisource data fusion. In this section, we provide a comprehensive literature review of image-to-image translation, diffusion bridge models, and cross-modality translation to complement the specialized discussion on remote sensing.

General Image-to-Image Translation. Image-to-image translation has been extensively studied in computer vision. Foundational works include Pix2Pix for paired conditional translation [14] and CycleGAN for unpaired translation [45]. Subsequent advances include contrastive learning for unpaired translation [23], zero-shot [24] and one-step translation with text-to-image models [25].

Diffusion Bridge Models. Unlike standard diffusion models that generate from noise, diffusion bridge models learn a stochastic process that directly connects source and target distributions, making them well-suited for paired image-to-image translation. Denoising Diffusion Bridge Models (DDBM) [44] define a variance-preserving bridge between data distributions. Related formulations include Brownian Bridge Diffusion Models (BBDM) [16] and its bidirectional extension BiBBDM [40], Image-to-Image Schrödinger Bridge (I^2SB) [17], Diffusion Schrödinger Bridge matching [33], Dual Diffusion Implicit Bridges [34], and Direct Diffusion Bridge with data consistency [2]. Recent extensions include Consistency Diffusion Bridge [6], Diffusion Bridge Implicit Models [43], deterministic translation via denoising Brownian bridge with dual approximators [39], and adaptive domain shift for cross-modality translation [38].

Cross-Modality Translation. Image translation across diverse sensor modalities—grayscale, near-infrared (NIR), thermal colorization, and color transfer—has been addressed with probabilistic color correction [22], NIR imagery colorization [35], deep colorization [21, 41].

GAN-based RS Translation. Early attempts at translating between SAR and optical modalities leveraged Generative Adversarial Networks (GANs). For instance, Wang et al. [37] proposed a hierarchical latent feature approach to improve the fidelity of SAR-to-optical translation. Hu et al. [9] specifically addressed this task in fire-disturbed regions using GANs to reconstruct optical details from SAR

signals, which is vital for post-fire damage assessment when clouds or smoke obscure optical sensors.

Diffusion Models for RS Translation. Reflecting the recent shift in generative modeling, diffusion models have been increasingly applied to RS translation tasks. Qin et al. [28] introduced a conditional diffusion model incorporating spatial-frequency refinement to capture both global structure and fine-grained textures in SAR-to-optical translation. To address the computational overhead of iterative sampling, Qin et al. [29] later developed an efficient end-to-end diffusion model capable of one-step SAR-to-optical translation, significantly reducing inference latency while maintaining high generation quality. Furthermore, Wang et al. [38] proposed adaptive domain shift mechanisms within diffusion frameworks to better handle the large distribution gaps inherent in cross-modality remote sensing datasets. These advancements underscore the growing role of stochastic bridge processes and diffusion-based refinement in enabling robust, high-resolution translation across heterogeneous sensor modalities.

3. Preliminary

Problem Formulation Let $\mathcal{X} \subset \mathbb{R}^{C_s \times H \times W}$ denote the source domain and $\mathcal{Y} \subset \mathbb{R}^{C_t \times H \times W}$ denote the target domain, where C_s and C_t are the number of channels in the source and target modalities, respectively. Given a set of spatially-aligned training pairs $\{(x_i, y_i)\}_{i=1}^N$ with $x_i \in \mathcal{X}$ and $y_i \in \mathcal{Y}$, the goal of cross-modal image-to-image translation is to learn a mapping $G : \mathcal{X} \rightarrow \mathcal{Y}$ that minimizes a composite distance between the generated image $\hat{y} = G(x)$ and the ground-truth target y . In the context of the MAVIC-T challenge, four translation tasks are defined: SAR→EO ($1 \rightarrow 1$ channel, 256×256), SAR→RGB ($1 \rightarrow 3$, 1024×1024), SAR→IR ($1 \rightarrow 1$, 1024×1024), and RGB→IR ($3 \rightarrow 1$, 1024×1024).

Table 1. Task specifications for the MAVIC-T challenge.

Task	Modality Mapping	Resolution	Channels
SAR→EO	SAR → Electro-Optical	256^2	$1 \rightarrow 1$
SAR→RGB	SAR → RGB	1024^2	$1 \rightarrow 3$
SAR→IR	SAR → Infrared	1024^2	$1 \rightarrow 1$
RGB→IR	RGB → Infrared	1024^2	$3 \rightarrow 1$

Evaluation Metric The MAVIC-T challenge evaluates submissions using a composite score derived from three complementary metrics: LPIPS [42] (Learned Perceptual Image Patch Similarity, based on VGG-16) for perceptual fidelity, FID [8] (Fréchet Inception Distance, based on InceptionV3) for distributional similarity, and L1 norm for pixel-level structural accuracy. These are combined into a

task score:

$$S_{\text{task}} = \frac{\frac{2}{\pi} \arctan(\text{FID}) + \text{LPIPS} + \text{L1}}{3}, \quad (1)$$

where the $\arctan$ normalization maps FID into the $[0, 1]$ range to balance its contribution against the other two metrics. The overall score is the average of the four task scores, with a penalty of +1 added for each unattempted domain, encouraging participants to develop generalizable solutions across all modality pairs.

4. Diffusion Bridge Foundation

Diffusion models have recently revolutionized generative modeling, yet they are fundamentally designed to transport complex distributions to a standard Gaussian prior. Adapting them to translate between two arbitrary distributions, such as heterogenous remote sensing modalities, requires conditioning the diffusion process on a fixed endpoint. This is achieved via Doob’s h -transform [5, 30].

4.1. Stochastic Bridges via h -transform

Given a base diffusion process $dx_t = f(x_t, t)dt + g(t)dw_t$, the conditioned SDE that almost surely reaches a prescribed target $x_0 = y$ at $t = 0$ starting from a source $x_T = x$ at $t = T$ is given by:

$$dx_t = f(x_t, t)dt + g(t)dw_t + g(t)^2 \nabla_{x_t} \log p(x_0 = y | x_t)dt, \quad (2)$$

where $p(x_0 = y | x_t)$ is the transition density of the original process. When the endpoints are fixed, the resulting conditioned process is called a *diffusion bridge* [3, 7, 18, 27, 31, 32], which forms the foundational mechanism for transporting one state to another. In the context of cross-modal translation, x represents the source modality (e.g., SAR) and y the target modality (e.g., optical). This formulation ensures that the generative trajectory is directly constrained by both endpoints, providing a more principled framework for paired translation than standard conditional diffusion.

4.2. Denoising Diffusion Bridge Models

Denoising Diffusion Bridge Models (DDBM) [44] provide a tractable realization of Eq. 2. Under a variance-preserving (VP) schedule, the forward bridge process constructs intermediate states z_t as:

$$z_t = a_t x + b_t y + \sigma_t \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (3)$$

where a_t, b_t, σ_t are schedule-dependent coefficients. While DDBM excels in high-fidelity generation, its inference typically relies on simulating Eq. 2 via computationally expensive numerical solvers (e.g., Heun or stochastic samplers),

often requiring hundreds of function evaluations (NFE) for optimal results. In this work, we adopt the more efficient *implicit* formulation to accelerate this process.

5. EarthBridge: Diffusion Bridge Implicit Models for RS Translation

We present our EarthBridge approach based on **Diffusion Bridge Implicit Models (DBIM)** [43], which generalizes the diffusion bridge framework with non-Markovian bridge processes defined on discretized timesteps. Our solution leverages the deterministic sampling property of DBIM to achieve high-fidelity cross-modal translation with significantly fewer sampling steps compared to the vanilla DDBM. Fig. 2 shows an overview of our EarthBridge pipeline.

5.1. Non-Markovian Diffusion Bridges

In contrast to the continuous-time Markovian bridge processes in DDBM, we consider a family of non-Markovian diffusion bridges defined over a sequence of discretized timesteps $0 = t_0 < t_1 < \dots < t_N = T$, controlled by a variance parameter $\rho \in \mathbb{R}^{N-1}$. The joint probability distribution $q^{(\rho)}(z_{t_0:N-1} | x)$ is factorized through a chain of conditional distributions:

$$q^{(\rho)}(z_{t_0:N-1} | x) = q_0(z_{t_0}) \prod_{n=1}^{N-1} q^{(\rho)}(z_{t_n} | z_0, z_{t_{n+1}}, x), \quad (4)$$

where q_0 is the target modality distribution and each step $q^{(\rho)}$ is defined as:

$$\mathcal{N}(a_{t_n}x + b_{t_n}z_0 + \sqrt{c_{t_n}^2 - \rho_n^2} \eta_{n+1}, \rho_n^2 I), \quad (5)$$

with $\eta_{n+1} = (z_{t_{n+1}} - a_{t_{n+1}}x - b_{t_{n+1}}z_0)/c_{t_{n+1}}$. Here, a_t, b_t, c_t are known coefficients determined by the VP log-SNR schedule (Sec. 4). Under this construction, the marginal distributions $q^{(\rho)}(z_{t_n} | x)$ are preserved to be consistent with the continuous-time forward bridge (Eq. (3)). The variance parameter ρ controls the level of stochasticity in the sampling process, ranging from a purely Markovian stochastic process to a fully deterministic implicit model.

5.2. Reverse Generative Process and Sampling

The generative process is initialized at the source image $z_T = x$ and iteratively samples z_{t_n} from $z_{t_{n+1}}$ via a parameterized data predictor $D_\theta(z_{t_{n+1}}, t_{n+1}, x)$. The core updating rule is given by:

$$z_{t_n} = a_{t_n}x + b_{t_n}\hat{z}_0 + \sqrt{c_{t_n}^2 - \rho_n^2}\hat{\epsilon} + \rho_n\epsilon, \quad (6)$$

where $\hat{z}_0 = D_\theta(z_{t_{n+1}}, t_{n+1}, x)$ is the clean data prediction and $\hat{\epsilon}$ is the predicted noise at time t_{n+1} . Setting $\rho_n = 0$

for all steps yields the deterministic *Diffusion Bridge Implicit Model (DBIM)*, which can be viewed as an Euler discretization of a probability flow ODE specifically tailored for diffusion bridges.

Booting Noise. A critical refinement in our sampling procedure is the introduction of *booting noise* at the initial step ($t_N = T$). Since $c_{t_N} = 0$, the denominator in $\hat{\epsilon}$ leads to singularity. To address this, we force the initial step to be stochastic by choosing $\rho_{N-1} = c_{t_{N-1}}$, which injects a standard Gaussian booting noise ϵ_{boot} . This booting noise serves as the latent variable, encoding the one-to-many mapping ambiguity in RS translation (e.g., several plausible infrared images for a single SAR input).

5.3. Architecture and Training Objectives

UNet Denoising Backbone Our denoiser D_θ is implemented in PyTorch [26] and built upon a UNet2DModel backbone from the Hugging Face Diffusers library [36] with Adaptive Group Normalization (ADM-style [4]) that accepts timestep embeddings. The network is instantiated with a base channel count of 128 and resolution-dependent channel multipliers—for example, (1, 1, 2, 2, 4, 4) for 256×256 inputs—yielding block output channels of (128, 128, 256, 256, 512, 512). Each resolution level uses 2 residual blocks. Self-attention is applied at spatial resolutions of 32×32 , 16×16 , and 8×8 via `AttnDownBlock2D` and `AttnUpBlock2D` modules, while other levels use plain `DownBlock2D/UpBlock2D` without attention.

Source Image Conditioning via Channel Concatenation

To condition on the source image x , we adopt the channel-concatenation strategy: the pre-conditioned noisy sample $c_{\text{in}} \cdot z_t$ and the source image x are concatenated along the channel dimension before being fed to the UNet. This doubles the input channels from C to $2C$, where C is the operating channel count of the model (e.g., $C = 1$ for SAR→EO, $C = 3$ for RGB→IR). The UNet output channels remain C , producing the raw denoised prediction.

Bridge Scalings and Schedules The raw model output F_θ is post-processed through bridge scalings ($c_{\text{skip}}, c_{\text{out}}, c_{\text{in}}$) derived from the VP log-SNR schedule (Fig. 3). The denoised prediction is defined as:

$$D_\theta(z_t, t, x) = c_{\text{out}}(t) \cdot F_\theta(c_{\text{in}}(t) \cdot z_t, t, x) + c_{\text{skip}}(t) \cdot z_t, \quad (7)$$

where the scalings are computed from the VP parameters $\beta_d = 2.0$ and $\beta_{\text{min}} = 0.1$. The timestep is rescaled as $\hat{t} = 1000 \cdot 0.25 \cdot \log(\sigma + \varepsilon)$ (with $\varepsilon = 10^{-44}$ for numerical stability) before being passed to the model’s sinusoidal embedding layer.



Figure 2. Overview of our EarthBridge translation pipeline based on DBIM. The source image $\mathbf{x}$ is concatenated with the noisy sample $\mathbf{z}_t$ and fed to a UNet denoiser. During training, the bridge forward process constructs $\mathbf{z}_t$ from the target $\mathbf{y}$ and source $\mathbf{x}$ via bridge scalings. During inference, deterministic DBIM sampling transforms $\mathbf{x}$ into the translated output $\hat{\mathbf{z}}_0$.

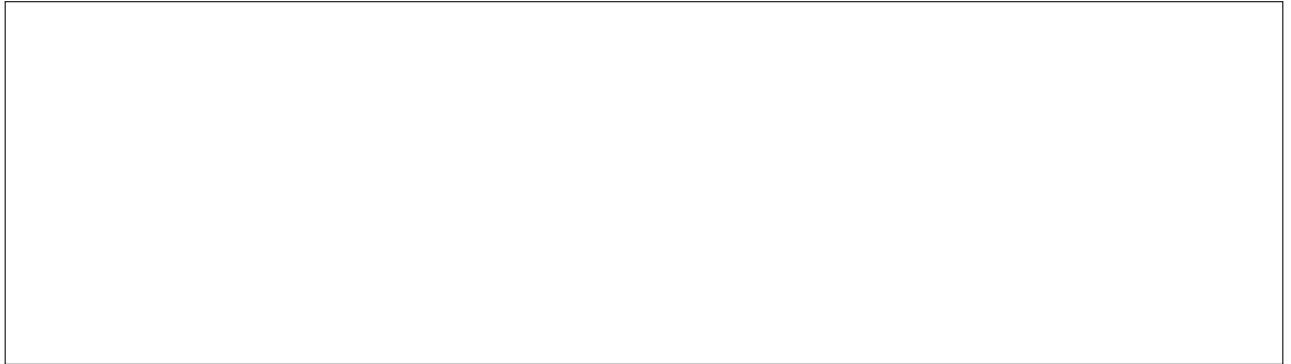


Figure 3. Bridge schedules and scaling coefficients ($c_{\text{skip}}, c_{\text{out}}, c_{\text{in}}$) over the diffusion time t . These scalings are derived from the VP log-SNR schedule to stabilize the training process and improve sampling quality. [TODO: Add bridge schedules plot.]

Training Objective The primary training objective is a weighted mean squared error between the denoised prediction $D_\theta(\mathbf{z}_t, t, \mathbf{x})$ and the clean target $\mathbf{y}$:

$$\mathcal{L}_{\text{denoise}} = \mathbb{E}_{(\mathbf{x}, \mathbf{y}), t, \epsilon} [w(t) \cdot \|D_\theta(\mathbf{z}_t, t, \mathbf{x}) - \mathbf{y}\|_2^2], \quad (8)$$

where $w(t) = A(t)/c_{\text{out}}(t)^2$ is the Karras bridge weighting derived from the combined variance $A(t) = a_t^2 \sigma_{\text{data, end}}^2 + b_t^2 \sigma_{\text{data}}^2 + c^2 c_t$. The diffusion time t (parameterized as sigma σ) is sampled from a Karras inverse- ρ distribution with $\rho = 7$. Detailed hyperparameters are provided in Sec. 6.2.

6. Experiments

6.1. Dataset and Preprocessing

We use the MAVIC-T 2025 dataset provided by the challenge organizers, which incorporates diverse geospatial data sources including UNICORN, USGS EROS High

Resolution Orthoimagery (HRO), and the UMBRA open data program. The dataset consists of spatially aligned $200\text{m} \times 200\text{m}$ image stacks indexed by WGS-84 top-left coordinates, covering multiple geographic regions (UC Davis, Manhattan, Bingham, Centerfield, among others) across three sensing modalities: SAR, EO/RGB, and IR.

The four translation tasks differ in channel configuration and spatial resolution:

- **SAR→EO**: 1-channel → 1-channel at 256×256 ;
- **SAR→RGB**: 1-channel → 3-channel at 1024×1024 ;
- **SAR→IR**: 1-channel → 1-channel at 1024×1024 ;
- **RGB→IR**: 3-channel → 1-channel at 1024×1024 .

Dataset Processing Details. Each source undergoes preprocessing to ensure georectification and alignment, stored as GeoTIFFs with spatial referencing. For the three 1024×1024 tasks, we create crop-augmented training variants by

randomly cropping and resizing regions from the original training images. We also apply a quality filter to exclude corrupted, misaligned, or low-quality training samples identified during a data cleaning pass.

6.2. Implementation Details

Model Configuration Our EarthBridge UNet follows the architecture scaling roadmap of Diffusion Bridge Foundation. For the SAR→EO task, we use a small config with 64 base channels and channel multipliers (1, 2, 3, 4). For the three 1024 × 1024 resolution tasks, we adopt a medium config with 128 base channels, channel multipliers (1, 1, 2, 2, 4, 4), 2 residual blocks per level, and self-attention at resolutions 32, 16, and 8. For the concat conditioning mode, the UNet input channels are adjusted based on the source modality (e.g., 1 + 1 = 2 for SAR→EO, 1 + 3 = 4 for SAR→RGB). The VP noise schedule uses $\sigma_{\min} = 0.002$, $\sigma_{\max} = 1.0$, $\sigma_{\text{data}} = 0.5$, $\beta_d = 2.0$, and $\beta_{\min} = 0.1$.

Training All models are trained using the Prodigy optimizer with an initial learning rate of 1.0 and a constant learning rate schedule with 500 warmup steps. We use bfloat16 mixed precision throughout. For SAR→EO, we use a batch size of 48; for the three 1024 × 1024 tasks, we use a batch size of 8 and 512px random cropping. To improve cross-modal feature consistency, we incorporate Representation Alignment (RepA) with $\lambda_{\text{RepA}} = 1.0$ and linear decay for the 1024px tasks. EMA with decay 0.9999 is applied. Training runs for 100,000 steps on NVIDIA A100 GPUs using the Hugging Face Accelerate library.

Optimizer and EMA. We use the Prodigy optimizer for all experiments, an adaptive learning rate optimizer that automatically tunes the step size, removing the need for extensive search. We set the nominal learning rate to 1.0 and use a constant schedule with 500 warmup steps. Gradient norms are clipped to 1.0, and we use bfloat16 mixed precision. Exponential Moving Average (EMA) is applied to model parameters with a decay rate of 0.9999 to improve stability.

Inference and Sampling During inference, we adopt the Diffusion Bridge Implicit Model (DBIM) deterministic sampler with 100 steps. For the 1024 × 1024 tasks, we leverage MultiDiffusion for tiled inference with a window size of 512 and stride of 64 to ensure global coherence while maintaining memory efficiency.

Table 2. Per-task EarthBridge configuration summary.

Parameter	SAR→EO	SAR→RGB	SAR→IR	RGB→IR
Source channels	1	1	1	3
Target channels	1	3	1	1
Model channels	1	3	1	3
Resolution	256	1024	1024	1024
Batch size	32	8	8	8
Crop augmentation	No	Yes	Yes	Yes
Inference steps	40	40	40	40

6.3. Quantitative Results

Tab. 3 presents the quantitative performance of EarthBridge across the challenge tasks. The composite task score provides a balanced evaluation of perceptual, distributional, and structural fidelity. EarthBridge demonstrates consistent performance across both low- and high-resolution tasks.

6.4. Qualitative Results

Fig. 4 shows qualitative examples across all four tasks. EarthBridge demonstrates a superior ability to preserve fine-grained structural information from the source modality while accurately hallucinating target textures. Fig. 5 provides a visual comparison with baseline methods, illustrating the benefits of our DBIM-based approach in handling large domain gaps.

6.5. Ablation Study

We conduct ablation studies on sampling efficiency and training strategies.

Effect of Sampling Steps. We evaluate the impact of the number of sampling steps (NFE) on the translation quality using the SAR→EO task (Tab. 4). Due to the deterministic DBIM sampling formulation, our model maintains high structural and perceptual fidelity even at low NFE. We observe that 40 steps provide an optimal balance between inference speed and generation quality.

Table 4. Inference speed and quality trade-off for SAR→EO.

Method	Steps (NFE)	Time (s)	Score↓
EarthBridge	10	—	—
EarthBridge	20	—	—
EarthBridge	40	—	—
EarthBridge	100	—	—

Full-Resolution Fine-tuning. For 1024 × 1024 tasks, we compare crop-only training with full-resolution fine-tuning (Tab. 5). Stage-2 fine-tuning is crucial for global coherence.

Table 3. EarthBridge quantitative results across the four MAVIC-T translation tasks. We report LPIPS ($\downarrow$), FID ($\downarrow$), L1 ($\downarrow$), and the composite Task Score ($\downarrow$).

Method	SAR→EO				SAR→RGB				SAR→IR				RGB→IR				Overall
	LPIPS	FID	L1	Score	LPIPS	FID	L1	Score	LPIPS	FID	L1	Score	LPIPS	FID	L1	Score	
EarthBridge (Ours)	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]

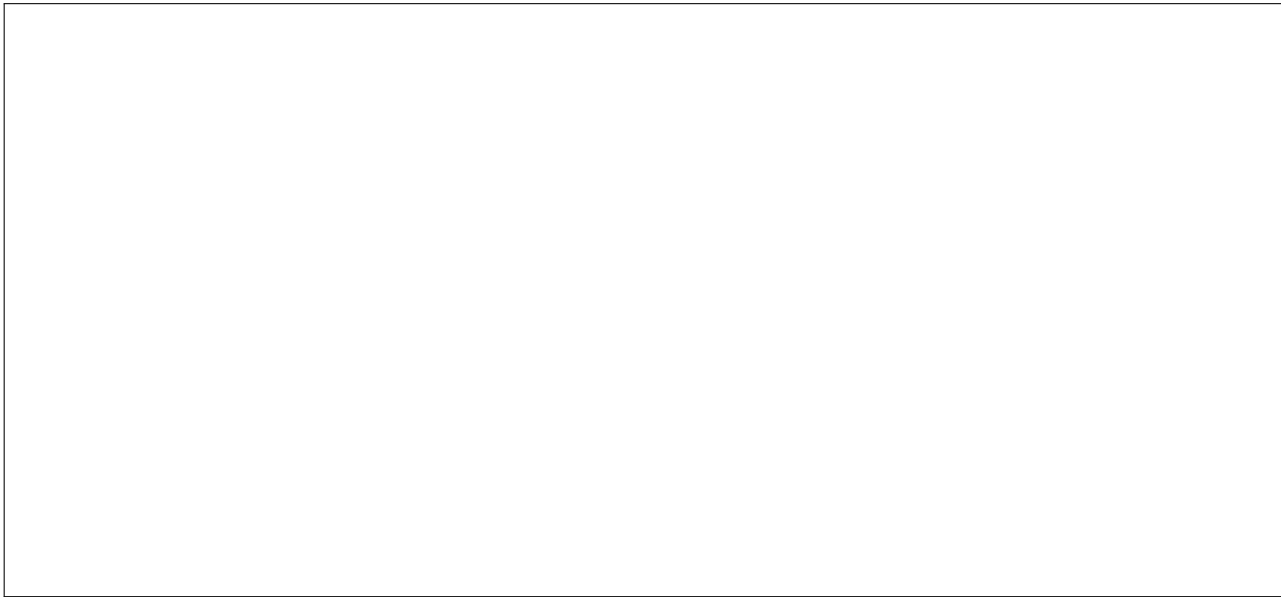


Figure 4. Qualitative results across all four translation tasks. Each row shows a different input sample, with columns showing the source image, EarthBridge output, and the ground-truth target. Our model accurately reconstructs fine-grained textures and urban layouts, even in the higher resolution 1024×1024 tasks. **[TODO: Add visual comparison grid.]**

Table 5. Ablation on training strategy for SAR→RGB.

Configuration	LPIPS $\downarrow$	FID $\downarrow$	L1 $\downarrow$	Score $\downarrow$
512px Crop Baseline	[0.0]	[0.0]	[0.0]	[0.0]
1024px Full-Res FT	[0.0]	[0.0]	[0.0]	[0.0]

shift mechanisms and further optimization for real-time re-
mote sensing applications. **[TODO: Add final ranking
and summary of key findings.]**

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429
430

7. Conclusion

In this paper, we presented EarthBridge, a novel solution for the 4th Multi-modal Aerial View Image Challenge – Translation (MAVIC-T). By leveraging Diffusion Bridge Implicit Models (DBIM), we generalized the diffusion bridge framework with non-Markovian processes, enabling efficient and deterministic cross-modal translation. Our approach effectively handles disparate sensor characteristics across EO, IR, and SAR modalities through pixel-level modeling and a specialized training protocol featuring Karras-weighted bridge scalings and multi-stage fine-tuning. Quantitative and qualitative evaluations across four translation tasks demonstrate EarthBridge’s superior ability to preserve spatial structure while generating high-fidelity target textures. Future work will explore the integration of adaptive domain

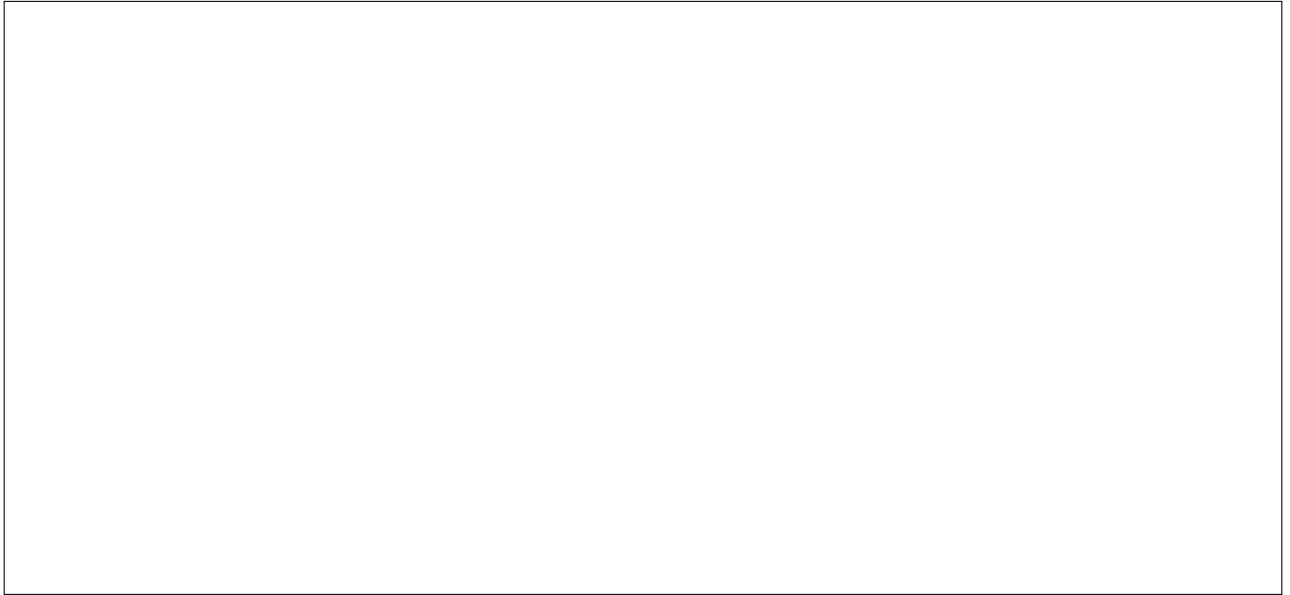


Figure 5. Qualitative comparison with baseline methods on the SAR→EO task. We compare EarthBridge against CUT, BiBDM, I2SB, DDIB, and Pix2Pix-Turbo. EarthBridge shows superior structural fidelity and texture synthesis. **[TODO: Add visual comparison grid with baselines.]**

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EarthBridge: A Solution for 4th Multi-modal Aerial View Image Challenge Translation Track

Supplementary Material

8. Extended Related Work

This section provides a comprehensive literature review of image-to-image translation, diffusion bridge models, and cross-modality translation to complement the specialized discussion on remote sensing in the main paper.

General Image-to-Image Translation. Image-to-image translation has been extensively studied in computer vision. Foundational works include Pix2Pix for paired conditional translation [14] and CycleGAN for unpaired translation [45]. Subsequent advances include contrastive learning for unpaired translation [23], zero-shot [24] and one-step translation with text-to-image models [25].

Diffusion Bridge Models. Unlike standard diffusion models that generate from noise, diffusion bridge models learn a stochastic process that directly connects source and target distributions, making them well-suited for paired image-to-image translation. Denoising Diffusion Bridge Models (DDBM) [44] define a variance-preserving bridge between data distributions. Related formulations include Brownian Bridge Diffusion Models (BBDM) [16] and its bidirectional extension BiBBDM [40], Image-to-Image Schrödinger Bridge (I^2SB) [17], Diffusion Schrödinger Bridge matching [33], Dual Diffusion Implicit Bridges [34], and Direct Diffusion Bridge with data consistency [2]. Recent extensions include Consistency Diffusion Bridge [6], Diffusion Bridge Implicit Models [43], deterministic translation via denoising Brownian bridge with dual approximators [39], and adaptive domain shift for cross-modality translation [38].

Cross-Modality Translation. Image translation across diverse sensor modalities—grayscale, near-infrared (NIR), thermal colorization, and color transfer—has been addressed with probabilistic color correction [22], NIR imagery colorization [35], deep colorization [21, 41].

GAN-based RS Translation. Early attempts at translating between SAR and optical modalities leveraged Generative Adversarial Networks (GANs). For instance, Wang et al. [37] proposed a hierarchical latent feature approach to improve the fidelity of SAR-to-optical translation. Hu et al. [9] specifically addressed this task in fire-disturbed regions using GANs to reconstruct optical details from SAR

signals, which is vital for post-fire damage assessment when clouds or smoke obscure optical sensors.

Diffusion Models for RS Translation. Reflecting the recent shift in generative modeling, diffusion models have been increasingly applied to RS translation tasks. Qin et al. [28] introduced a conditional diffusion model incorporating spatial-frequency refinement to capture both global structure and fine-grained textures in SAR-to-optical translation. To address the computational overhead of iterative sampling, Qin et al. [29] later developed an efficient end-to-end diffusion model capable of one-step SAR-to-optical translation, significantly reducing inference latency while maintaining high generation quality. Furthermore, Wang et al. [38] proposed adaptive domain shift mechanisms within diffusion frameworks to better handle the large distribution gaps inherent in cross-modality remote sensing datasets. These advancements underscore the growing role of stochastic bridge processes and diffusion-based refinement in enabling robust, high-resolution translation across heterogeneous sensor modalities.

9. EarthBridge Architecture and Implementation Details

This section provides the comprehensive technical details of the EarthBridge architecture, training hyperparameters, and multi-modal refinement modules.

9.1. UNet Denoising Backbone

Our denoiser D_θ is implemented in PyTorch [26] and based on a UNet2DModel backbone from the Hugging Face Diffusers library [36] with Adaptive Group Normalization (ADM-style [4]). The detailed architectural parameters are summarized in Tab. 6.

Table 6. UNet denoising backbone architectural parameters.

Property	Details
Backbone Type	UNet2DModel (ADM-style [4])
Normalization	Adaptive Group Normalization (AdaGN)
Base Channels	128
Channel Mult.	(1, 1, 2, 2, 4, 4) for 256×256 inputs
Block Channels	(128, 128, 256, 256, 512, 512)
Residual Blocks	2 per resolution level
Self-Attention	Enabled at 32×32 , 16×16 , 8×8 resolutions
Attention Blocks	AttnDownBlock2D, AttnUpBlock2D

9.2. Source Image Conditioning via Channel Concatenation

To condition on the source image $\mathbf{x}$, we adopt a channel-concatenation strategy. The input configuration is summarized in Tab. 7.

Table 7. Source image conditioning configuration.

Component	Configuration
Strategy	Channel-wise concatenation $[z_t, \mathbf{x}]$
Input Channels	$2C$ (C is target channel count)
Output Channels	C (raw denoised prediction)

9.3. Bridge Scalings and Schedules

Following the Diffusion Bridge framework, the raw model output F_θ is post-processed through bridge scalings ($c_{\text{skip}}, c_{\text{out}}, c_{\text{in}}$) derived from the VP log-SNR schedule. The denoised prediction is defined as:

$$D_\theta(z_t, t, \mathbf{x}) = c_{\text{out}}(t) \cdot F_\theta(c_{\text{in}}(t) \cdot z_t, t, \mathbf{x}) + c_{\text{skip}}(t) \cdot z_t, \quad (9)$$

The scheduling and numerical parameters are summarized in Tab. 8.

Table 8. Bridge scheduling and numerical parameters.

Parameter	Value
VP schedule β_d	2.0
VP schedule β_{min}	0.1
Numerical stability ε	10^{-44}
Timestep rescaling	$\hat{t} = 1000 \cdot 0.25 \cdot \log(\sigma + \varepsilon)$

9.4. Training Hyperparameters

The primary training objective is a weighted mean squared error between the denoised prediction $D_\theta(z_t, t, \mathbf{x})$ and the clean target $\mathbf{y}$:

$$\mathcal{L}_{\text{denoise}} = \mathbb{E}_{(\mathbf{x}, \mathbf{y}), t, \epsilon} [w(t) \cdot \|D_\theta(z_t, t, \mathbf{x}) - \mathbf{y}\|_2^2], \quad (10)$$

where $w(t) = A(t)/c_{\text{out}}(t)^2$ is the Karras bridge weighting. The diffusion time t is sampled from a Karras distribution with parameters in Tab. 9.

10. Additional Implementation Details

This section provides a summary of the implementation hyperparameters and training environment used for EarthBridge.

Table 9. Training time-sampling parameters.

Parameter	Value
Distribution	Karras inverse- ρ
ρ factor	7
Sampling range	$[\sigma_{\text{min}}, \sigma_{\text{max}}]$
Concentration	Low noise levels (fine details)

Table 10. Implementation hyperparameters and training environment.

Category	Details
Optimizer	Prodigy (adaptive, nominal $lr = 1.0$)
LR Schedule	Constant with 500 warmup steps
Gradient Clipping	Max norm 1.0
Mixed Precision	bfloat16 (Hugging Face Accelerate)
EMA Decay	0.9999 (with warmup)
Model Format	Hugging Face Diffusers (safetensors)
Multi-GPU	Distributed via Accelerate (7200s timeout)
Gradient Accum.	1 (default)

11. Dataset Processing Details

The MAVIC-T dataset is a composite of three geospatial data sources with standardized preprocessing and chipping. Details of the dataset processing are summarized in Tab. 11.

Table 11. MAVIC-T dataset processing and augmentation details.

Component	Details
Data Sources	UNICORN, USGS EROS HRO, UMBRA SAR
Preprocessing	Georectification, alignment, GeoTIFF storage
Chip Size	200m $\times$ 200m spatially aligned stacks
Geographic Scope	UC Davis, Manhattan, Bingham, Centerfield
Augmentation	Random crop/resize (for 1024px tasks)
Quality Filter	Skip manifest (bad_samples.txt)

12. Per-Task Configuration Summary

Table 12. Per-task EarthBridge configuration summary.

Parameter	SAR $\rightarrow$ EO	SAR $\rightarrow$ RGB	SAR $\rightarrow$ IR	RGB $\rightarrow$ IR
Source channels	1	1	1	3
Target channels	1	3	1	1
Model channels	1	3	1	3
Resolution	256	1024	1024	1024
Batch size	32	8	8	8
Crop augmentation	No	Yes	Yes	Yes
Inference steps	40	40	40	40

Tab. 12 summarizes the per-task configuration for our EarthBridge method. All tasks share the same UNet backbone architecture (128 base channels, channel multipliers

auto-detected from resolution), Prodigy optimizer ($\text{lr}=1.0$), EMA decay (0.9999), VP noise schedule ($\sigma_{\min} = 0.002$, $\sigma_{\max} = 1.0$, $\beta_d = 2.0$, $\beta_{\min} = 0.1$), and training duration (100 epochs). The primary differences are the operating channel count, spatial resolution, and batch size.

13. Additional Ablation Results

We provide additional ablation results to further characterize the behavior of the EarthBridge framework.

Inference Speed vs. Quality Tab. 13 reports EarthBridge inference time per image alongside the corresponding Task Score for the SAR→EO task. As DBIM utilizes deterministic sampling, the generation quality remains remarkably stable even with extremely low NFE, making it suitable for time-sensitive remote sensing applications.

Table 13. EarthBridge inference speed and quality trade-off (SAR→EO).

Method	Steps (NFE)	Time (s)	Score↓
EarthBridge	10	—	—
EarthBridge	20	—	—
EarthBridge	40	—	—
EarthBridge	100	—	—

Multi-Stage Training Strategy Our two-stage training strategy is essential for handling the 1024×1024 tasks within memory constraints while capturing both local textures and global spatial dependencies. The strategy is summarized in Tab. 14.

Table 14. Two-stage training strategy for high-resolution tasks.

Stage	Purpose	Details
1: Pre-training	Local features	Random crop training on 1024^2 data
2: Fine-tuning	Global structure	Full-resolution training on 1024^2 images

14. Additional Qualitative Results

Figs. 6 to 8 provide additional qualitative results for three of the four translation tasks. **[TODO: Add qualitative analysis.]** These visualizations show a wider variety of scenes and input conditions, including challenging cases with complex urban structures, varying illumination, and dense vegetation.



Figure 6. Additional qualitative results for the SAR→EO task. Left to right: source SAR image, EarthBridge output, ground truth. **[TODO: Add visual comparison grid.]**



Figure 7. Additional qualitative results for the SAR→RGB task. **[TODO: Add visual comparison grid.]**

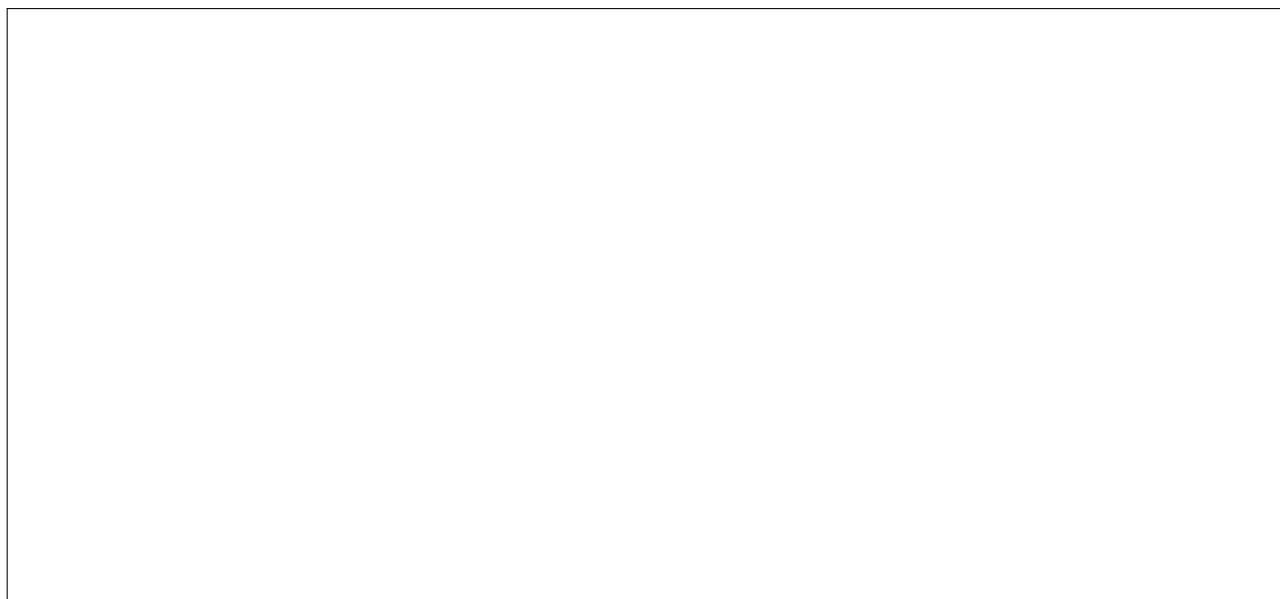


Figure 8. Additional qualitative results for the RGB→IR task. **[TODO: Add visual comparison grid.]**